FLASH: A Bit-Exact Integer CNN for Offline Chest X-Ray Pneumonia Screening on a Low-Cost Zynq-7020

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***Abstract*—FLASH is a pneumonia screener whose complete convolutional neural network runs in the programmable logic of a low-cost Zynq-7020 (PYNQ-Z2 board) and reproduces an independent integer reference model bit for bit. The network is trained directly in integer units on 224×224 chest radiographs from the RSNA Pneumonia Detection dataset. It has five stride-2 3×3 convolutions, global average pooling and a 64→2 classifier, with 47,602 parameters and 12.19 million multiply-accumulates (MACs) per image. A single convolution engine, configured at run time by a 128-bit descriptor per layer, executes the same Verilog at 28×28 and 224×224. Three bit-exact links connect the trained network, a NumPy golden model, RTL simulation and the board. On the board, all 244 verification images match the reference in logit0, logit1, margin and decision, on three bitstreams. At 66.67 MHz an image takes 182.94 ms, 99.97% of the engine's one-MAC-per-clock peak. On 4,003 held-out patients the model reaches an AUROC of 0.8229 (95% CI 0.8083–0.8379). The first board run produced no bit-exact result (0/244) although 234 of 244 decisions agreed with the reference. We located the cause offline from the board's logits, a data-width mismatch in the processor's memory bridge, and confirmed it by changing one register bit, which restored 244/244. A subgroup analysis exposes two failure modes: low specificity on radiographs with other abnormalities, and partial reliance on the radiographic projection. FLASH is a research prototype and is not a medical device.**

***Index Terms*—FPGA, CNN accelerator, integer inference, chest X-ray, pneumonia, verification, medical AI**

I. Introduction

Pneumonia killed 740,180 children under five in 2019, 14% of all deaths in that age group [1]. A chest radiograph is the usual confirmation, but it needs a reader, and where readers are scarce a film can wait a long time for its report. Deep networks can read chest radiographs [2], [3], but they usually run on a server, which needs connectivity and sends images off site. A small device that reads the film where it is taken, without a network connection, and gives an immediate first opinion could help a clinician decide which films to read first.

We use an FPGA because the circuit computes the same integers as the validated reference model on every run, without an operating system or a floating-point library in between. The hardware and the reference can therefore be compared exactly, output by output, and an integration fault appears as a difference in a 32-bit integer.

This paper describes FLASH V1, a research prototype on the PYNQ-Z2 board, and makes four contributions:

1) A bit-exact integer co-design and verification flow, from training to golden model to RTL to board, at 224×224 on a low-cost Zynq-7020 (Sections III and V).
2) A resolution-agnostic convolution engine driven by a layer table, which ran 28×28 and 224×224 with the same Verilog (Section IV).
3) A board-only integration fault that was localised offline from the logits and confirmed by intervention. Decision-level agreement was 234/244 while no logit was correct, so only the logit-level comparison detected it (Section VIII).
4) A subgroup analysis exposing two failure modes: confusion of other lung abnormalities ("Not Normal") with pneumonia, and a shortcut through the radiographic projection, AP or PA (Sections VII and IX).

II. Related Work

Chest radiograph datasets and models. ChestX-ray8 [2] released hospital-scale radiographs with text-mined labels, and CheXNet [3] trained a 121-layer network on them for pneumonia. The RSNA Pneumonia Detection dataset [4] added radiologist annotations of possible pneumonia to a subset, with a three-way label that separates "Normal" from "No Lung Opacity / Not Normal". Kermany *et al.* [5] released paediatric radiographs, and MedMNIST v2 [6] includes a 28×28 version of them (PneumoniaMNIST), which our earlier prototype used.

Shortcut learning. Zech *et al.* [7] showed that pneumonia models generalise unevenly across hospital systems and can learn to recognise the source hospital. DeGrave *et al.* [8] found COVID-19 radiograph classifiers relying on shortcuts instead of pathology, and Geirhos *et al.* [9] describe the general phenomenon.

Integer networks. Jacob *et al.* [10] established integer-arithmetic-only inference with quantization-aware training. FLASH trains directly in integer units, so the exported integers are the model itself. Global average pooling comes from Network in Network [11], and the replacement of pooling by strided convolution from the all-convolutional net [12].

FPGA CNN accelerators. Zhang *et al.* [13] used a roofline model to size compute and memory for CNN accelerators, Qiu *et al.* [14] mapped large CNNs to an embedded Zynq with dynamic-precision quantization, FINN [15] generates dataflow accelerators for binarized networks, and hls4ml [16] compiles

TABLE I
NETWORK AT 224×224

Layer	Output	Params[D]	MACs/image[D]
conv1, 1→8	8×112×112	80	903,168
conv2, 8→16	16×56×56	1,168	3,612,672
conv3, 16→32	32×28×28	4,640	3,612,672
conv4, 32→48	48×14×14	13,872	2,709,504
conv5, 48→64	64×7×7	27,712	1,354,752
GAP	64	0	(3,136 adds)
FC, 64→2	2 logits	130	128
Total		47,602	12,192,896

small networks to low-latency HLS designs. Sze *et al.* [17] survey the metrics. These works maximise throughput with many parallel processing elements, on other networks and resolutions, so a numerical comparison with FLASH would not be meaningful. FLASH uses one MAC per clock by design. Its aim is verifiable exactness at clinical resolution on a low-cost device, and parallelism is left to future work (Section X).

Reporting and regulation. CLAIM [18] and TRIPOD+AI [19] set reporting standards for imaging AI and prediction models. We use Wilson intervals [20] for proportions, the bootstrap [21] for AUROC, and the Hanley–McNeil standard error [22] where only an AUROC and class counts are available. Section IX relates FLASH to IEC 62304 [23], ISO 14971 [24], ISO 13485 [25] and the FDA's AI/ML action plan [26].

III. MODEL AND INTEGER CO-DESIGN

Data. V1 uses the RSNA dataset [4], [27]: 26,684 adult patients, one frontal radiograph each, stored as DICOM [28]. We split by patient into 18,678 training, 4,003 validation and 4,003 test patients; test prevalence is 22.5%. Preprocessing applies the modality look-up table and the window, inverts MONOCHROME1 images, letterboxes to a square, area-resizes to 224×224 and rounds to 8 bits. It passes seven self-tests, including real scanner files (JPEG 2000, MONOCHROME1, modality LUT before window).

Network. Five 3×3 convolutions with stride 2 and padding 1 take the channels 1→8→16→32→48→64. Global average pooling (GAP) reduces each of the 64 final maps to one value, and a dense layer maps them to two logits (Table I).[1] Because GAP removes the spatial size, the classifier and the parameter count do not depend on the input resolution.

Integer rules. Training, the golden model and the hardware share one arithmetic. Pixels and activations are unsigned 8-bit, weights signed 8-bit in $[-127, 127]$, biases 32-bit. A convolution output is the bias plus the sum of weight×pixel over the 3×3 window and all input channels. It is shifted right by a per-layer amount (7, 7, 7, 8, 8) and clamped to 0..255, and the clamp acts as the ReLU. GAP sums each 7×7 map and shifts right by 6, which divides by 64 instead of 49 and needs no divider. The decision uses the margin $m = \text{logit}_1 - \text{logit}_0$, and an image is flagged when $m > T$.

Accumulator sizing. The accumulator width comes from the trained weights. For each output channel, $\sum \max(w, 0) \cdot 255 + b$ and $\sum \min(w, 0) \cdot 255 + b$ bound the sum for any 8-bit input image. The widest layer needs 22 bits, and the margin 19 bits. The range observed on 8,006 validation and test images lies well inside that bound. Because an overflow would produce a wrong result without any error, the width is set by this worst-case bound.

Training. The weights are trained directly in integer units, first unconstrained (24 epochs), then with rounding on every forward pass and a straight-through gradient (16 epochs). The epoch with the best validation AUROC is kept (0.8380). The threshold $T = -646$ was chosen on the validation set for at least 90% sensitivity and locked before the test set was examined. On the 4,003 test patients the model reaches an AUROC of 0.8229 (bootstrap 95% CI 0.8083–0.8379), with sensitivity 91.0% (89.0–92.7) and specificity 53.9% (52.1–55.6).

IV. HARDWARE ARCHITECTURE

Figure 1 shows the system. The ARM core copies an image into a DMA buffer, writes the threshold and a start bit, starts an AXI DMA [29] in simple mode, polls a status register, and reads back two logits, the margin, the decision and a cycle count.

The core. `layer_seq` walks a ROM of layer descriptors, one 128-bit word per layer. Each holds the operation, kernel, stride and padding (4 bits each); input and output channels, heights and widths (12 bits each); the shift (6 bits); and the weight and bias base addresses (20 and 10 bits). It starts the right unit, swaps the two feature-map buffers between layers and streams the GAP input. `conv_engine` is shared by all five convolutions. It is pipelined as operand multiplexer, register, multiplier, register and accumulator, and does one MAC per clock. Padding is implemented as a gated read: an out-of-image tap contributes zero, so no padded frame is stored. `gap_unit` keeps 64 running sums, `fc_unit` computes two 32-bit logits, and `decision` compares the margin with a run-time threshold register.

Scaling from 28×28 to 224×224. The audited predecessor (v0) was a fixed 28×28 design with one hand-written module per stage. Its classifier flattened a 4×14×14 map into a 784→16 dense layer, so 99.7% of its 12,634 parameters sat in layers whose size grows with the image. At 224×224 its first dense layer alone would need 802,816 weights. In V1, moving from 28×28 (stage V1.1) to 224×224 (V1.2) needed only new memory files (weights, biases, layer table) and one RTL change: the feature-map buffers grew from 64 KB to 128 KB, because conv1's output is 8×112×112 = 100,352 bytes. The descriptor layout is identical at both sizes. MACs per image grow from 227,360 to 12,192,896, a factor of 53.6. That is less than the factor of 64 in pixels, because the small maps at 28×28 round up (7→4→2→1). In synthesis of the core alone, block RAM grows from 48.5 to 80.5 tiles. The convolution's

[1] In the tables, a superscript marks the origin of each value: M for values measured by a tool or a board run on the final design, S for Vivado estimates (all power figures), and D for values derived from measured ones. The repository recomputes every D value from the raw evidence.

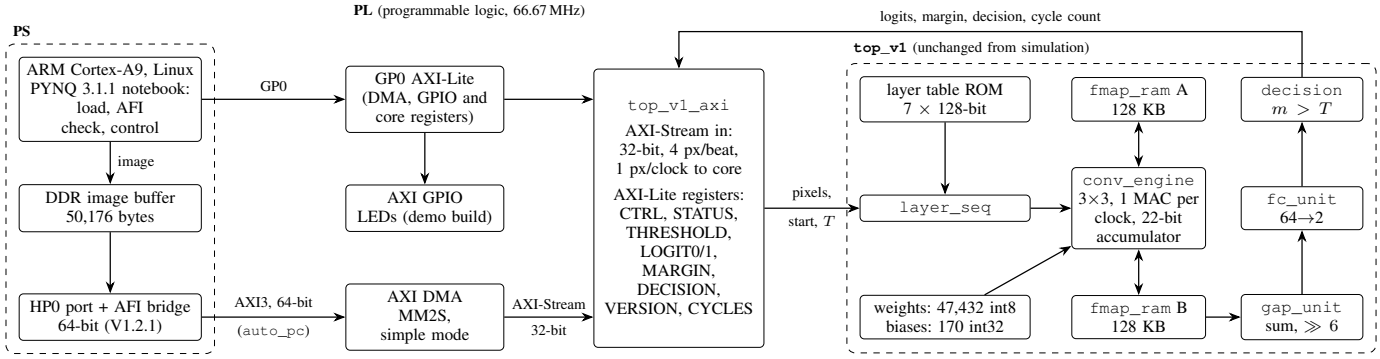


Fig. 1. FLASH V1 on the Zynq-7020. The ARM core only moves data and polls registers, and all inference runs in the programmable logic (PL). The input lands in buffer A, the layers alternate between A and B, and GAP reads conv5's output from B. V1.2.1 runs the HP0 bridge and the DMA's memory side at 64 bits (Section VIII).

address arithmetic ($c \cdot h \cdot w + y \cdot w + x$, a DSP multiply-add into the block-RAM address) is the critical path at both sizes.

The AXI wrapper. `top_v1_axi` accepts the image as 32-bit AXI-Stream beats, four pixels per beat, 12,544 beats per image, and feeds the core one pixel per clock. It reproduces the exact drive sequence of the simulation testbench: threshold, all 50,176 pixels, one idle cycle, one start pulse. It exposes the registers of Fig. 1, with VERSION = 0xF1A50102 and CYCLES counting core clocks from start to result.

V. Verification Methodology

Correctness is established by three links, each an exact comparison of integers with no tolerance.

1) *Trained network = golden model.* The golden model is an independent NumPy program in int64 that reads only the exported files. It gives logits identical to the training framework on all 4,003 test images, with 0 mismatches on every intermediate tensor of the 244 verification images.
2) *Golden model = RTL.* The simulator dumps every layer's output and a script compares it with the golden model, so a mismatch points to one layer. At 224×224, 16 of 16 images and 12 of 12 per-layer traces are exact in Vivado xsim. The AXI wrapper passes a 33-check testbench with varied bus timing and error cases. An independent audit re-ran the golden model against all 244 vectors and 60 traces of both stages, with 0 mismatches.
3) *RTL = board.* The board compares logit0, logit1, margin and decision with the expected files on all 244 images (Section VII).

Unbiased vectors. The 244 verification images are a seeded, balanced slice of the test split, 122 pneumonia and 122 not. They were drawn without looking at the model's confidence, and the model misclassifies 70 of them at T. The hardware has to reproduce these errors as well, and an exact match of 32-bit logits on such images is very unlikely to occur by chance. The hackathon prototype from which FLASH descends was tested by comparing one decision bit on eight images that the model classified confidently, and three datapath defects remained undetected across its five versions.

TABLE II
Technical defects found, all phases

#	Defect	Found by
1–5	v0 (28×28): line buffer read a pixel before it was written; windows centred one row early; max-pool sampled block RAM a clock early; dense-layer pointer off by one; testbench drove inputs on the sampling edge	golden-model tensor diff
6	`gap_unit` testbench raced the DUT	isolated reproduction
7	conv and FC assumed 2-cycle reads; RAM had 1	integration sweep
8	sequencer needed two start pulses	integration sweep
9	64 KB buffer too small for conv1 at 224	first 224 simulation
10	testbench timeout too short; literal wrapped	simulation
11	synthesis loaded the 28×28 weights through RTL defaults	synthesis log audit
14	wrapper testbench sent a beat twice	first xsim run
15	HP0 bridge 64-bit, design 32-bit	board, then offline model search

Defect ledger. Table II lists the technical defects found in all phases, numbered as in the project's ledger. Defects 11 and 15 would have produced a device that gives wrong answers without any error message. Two process problems are not in the table: an early version of the documentation overstated the test coverage (12), and the export manifest records hashes for only 3 of the 313 memory files of each stage (13, still open).

VI. Implementation

The design targets the xc7z020clg400-1 with Vivado 2022.2. The block design holds the processing system, the AXI DMA (read channel only), `top_v1_axi` as a module reference, a reset block, and in one variant an AXI GPIO for the board LEDs. The DMA reaches DDR through the HP0 port. Vivado does not connect the AXI4 DMA directly to the AXI3 port, so a protocol converter sits between them.

Clock. The fabric clock is 66.67 MHz (1000 MHz/15). The processing system cannot synthesise 75 MHz exactly (a 75 MHz request gives 76.92 MHz), and the next lower frequency, 71.43 MHz, failed post-route timing with WNS

TABLE III
POST-ROUTE RESULTS, XC7Z020 AT 66.67 MHZ

	V1.2 (HP0 32-bit)	V1.2.1 (HP0 64-bit)	V1.2.1-LED
WNS / WHS (ns)[M]	+0.206/+0.024	+0.314/+0.030	+0.344/+0.025
Failing endpoints[M]	0/15,320	0/15,451	0/15,704
LUT[M]	3,818	3,864 (7.3%)	3,947
Flip-flops[M]	4,745	4,766 (4.5%)	4,899
BRAM tiles[M]	81.5	82 (58.6%)	82
DSP slices[M]	14	14 (6.4%)	14
Power, total (W)[S]	1.492	1.494	1.499
of which PS7 (W)[S]	1.256	1.256	1.256

TABLE IV
BOARD RUNS, 244 IMAGES EACH

Run	Bitstream	AFI bit 0	Bit-exact[M]	TP/FN/TN/FP[M]
1	V1.2.1	0	244/244	117/5/57/65
2	V1.2 (32-bit)	0	0/244	118/4/56/66
3	V1.2 (32-bit)	0→1	244/244	117/5/57/65
demo	V1.2.1-LED	0	244/244	117/5/57/65

−0.925 ns. At 66.67 MHz the critical path, from the convolution's address register through two DSP slices and one LUT into a block-RAM address pin, takes 13.786 ns of the 15 ns budget, 70% of it logic. The RTL was frozen for the board work, so we accepted the lower clock; pipelining that path is a V2 item.

Table III gives the post-route results. V1.2.1 differs from V1.2 only in the HP0 and DMA data width, and the LED variant adds the GPIO. The accelerator's netlist is identical in all three: its 2,274 memory-initialisation lines match.

Block RAM, mostly the two 128 KB feature-map buffers, is the only substantial resource. Of the 3,864 LUTs of V1.2.1, the accelerator uses 2,758. Power is a vector-less Vivado estimate (confidence Medium for the first two builds and Low for the LED build), and 84% of it is the processing system.

VII. BOARD EVALUATION

The board runs PYNQ 3.1.1 [30]. A notebook loads the bitstream, checks the version register, the fabric clock and the AFI setting, then runs all 244 verification images and writes a CSV file and a summary to the evidence folder.

Bit-exactness. Table IV lists the runs of 2026-10-09. On every image of every run, CYCLES was 12,196,126 and the error bit was never set. Runs 1 and 3 wrote byte-identical CSV files, and run 2 equals the failing board run of the previous day image by image.

Accelerator metrics. Table V gives the usual figures of merit. The 12,192,896 MACs take 12,196,126 cycles. The remaining 3,230 cycles are the GAP stream (3,136) and about 94 cycles of sequencing and pipeline drain, so the engine is busy with MACs in 99.97% of its cycles. With one MAC per clock the absolute performance is low. The FPGA is 3.6 to 4.1 times faster than the same NumPy code on the board's ARM core, but a desktop PC runs that code in 8.7 ms. The ARM baseline is unoptimised int64 NumPy on one core.

TABLE V
ACCELERATOR METRICS, V1.2.1 AT 66.67 MHZ. PL POWER IS THE CHIP ESTIMATE MINUS THE PS7 SHARE

Latency, compute[M]	182.94 ms
Latency, end to end[M]	184.73 ms
Throughput, compute / end to end[D]	5.47 / 5.41 img/s
Work per image (2 ops/MAC)[D]	24.39 MOP
Achieved / peak (1 MAC/clock)[D]	0.13330 / 0.13333 GOP/s
MAC-cycle efficiency[D]	99.97%
DSP / BRAM tiles / LUT[M]	14 / 82 / 3,864
Power, chip / PL[S]	1.494 / 0.238 W
Energy per image, chip / PL[D]	273.3 / 43.5 mJ
Efficiency, chip / PL[D]	0.089 / 0.56 GOP/s/W
PC (i7-13650HX), NumPy int64[M]	8.7 ms
Board ARM Cortex-A9, same code[M]	665–744 ms
Board FPGA, compute / end to end[M]	182.94 / 184.73 ms

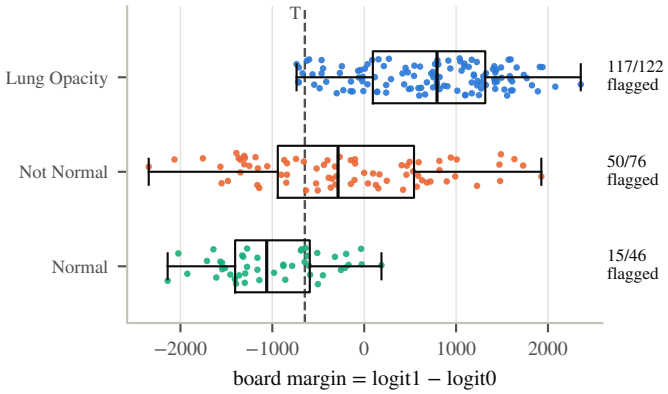


Fig. 2. Board margins on the 244 images by RSNA class (measured, board run 1). Dots are images, boxes show the quartiles and the median, and the dashed line is $T = -646$. Two thirds of the "Not Normal" negatives lie above T.

Classification on the 244. The board's own outputs give a sensitivity of 0.959 (Wilson 0.908–0.982), a specificity of 0.467 (0.381–0.555), PPV 0.643, NPV 0.919 and an AUROC of 0.837 (bootstrap 0.785–0.884, 2,000 resamples), consistent with the test-set AUROC. Of the 70 errors, 65 are false positives and 5 are false negatives, which matches a threshold chosen for at least 90% sensitivity. The five false negatives have margins from -736 to -670, all within 90 of T.

Figure 2 shows where the false positives come from. RSNA's negative class mixes "Normal" with "No Lung Opacity / Not Normal". Of the 76 Not-Normal negatives, 50 are flagged (65.8%), against 15 of the 46 Normal ones (32.6%). Their median margins, -284.5 and -1061, lie on opposite sides of T. Figure 3 shows the same pattern on the full test set, where pneumonia against Normal scores 0.936 and against Not Normal 0.738.

The subgroup analysis also exposes a shortcut. The projection alone, AP (taken at the bedside, often of sicker patients) or PA, predicts the label with an AUROC of 0.706. Within AP films or within PA films the model scores only 0.743 and 0.766, so part of its overall 0.823 comes from the projection. Men and women score the same within error (0.821 and

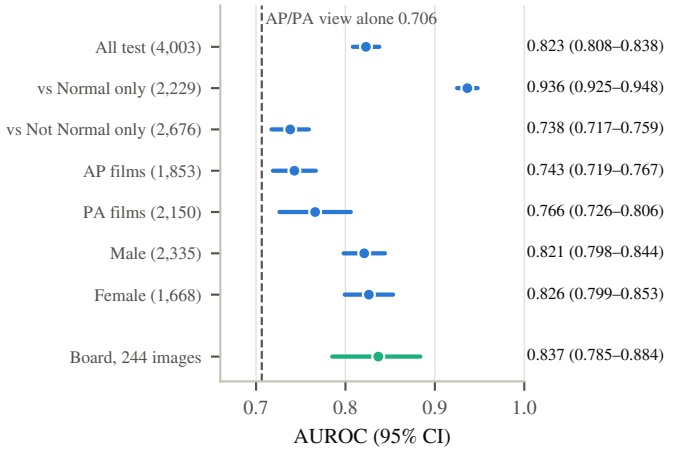


Fig. 3. Test-set AUROC by subgroup (measured, manifest of the training run). Intervals are the bootstrap for all test patients and for the board's 244 images, and Hanley–McNeil [22] otherwise (derived), which overall gives a slightly wider 0.805–0.840. The dashed line is the AUROC of the radiographic projection alone.

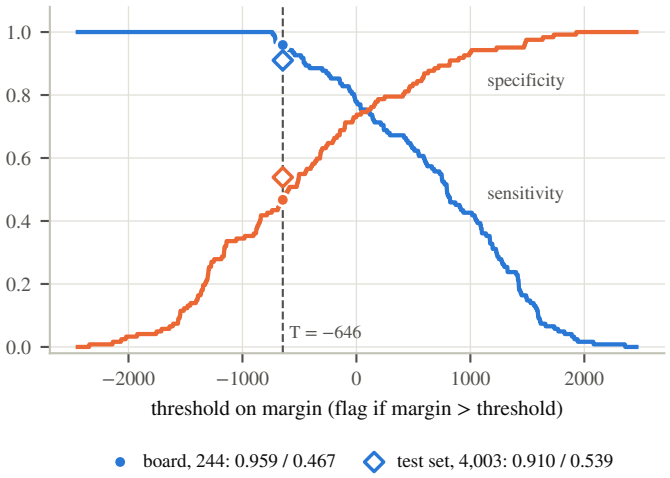


Fig. 4. Sensitivity and specificity against the threshold, from the 244 board margins (derived). Diamonds mark the test-set operating point. The plot is illustrative only, because T was fixed on validation data.

0.826).

Figure 4 shows the trade-off that a different T would give, and Figure 5 shows how a positive result should be read in deployment. At the test prevalence of 22.5% the PPV is 36.5% and the NPV 95.4%. At a prevalence of 5% the PPV falls to 9.4%, about one positive in eleven, while the NPV rises to 99.1%. The 244 subset is balanced by construction, so its PPV of 0.643 does not describe deployment. A negative result is therefore informative at realistic prevalences, whereas a positive result is best used to prioritise a radiograph for reading.

VIII. Case Study: A Fault Only the Board Could Show

Symptom. The first V1.2 bitstream produced no bit-exact result (0/244). The other indicators were normal (Fig. 6). The

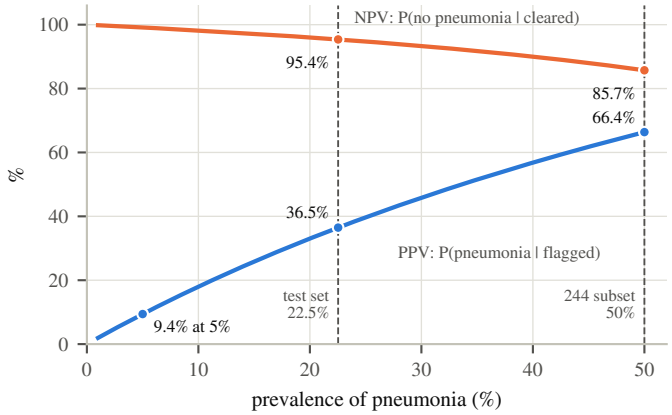


Fig. 5. PPV and NPV at the test operating point (821 of 902 positives flagged, 1,671 of 3,101 negatives cleared) as the prevalence varies (derived).

logits correlated with the expected values at 0.98, and 234 of 244 decisions agreed. The results were identical at 25, 50 and 66.67 MHz, which excludes a timing failure, and repeated runs and soft resets gave identical outputs. The DMA reported no error. A test that compared only decisions would have accepted this hardware.

Offline localisation. Instead of modifying the RTL, we used the golden model to interpret synthetic probe images sent to the board. Constant images and an image constant along each row came back exact, while a ramp along each row did not, which pointed at the order of pixels within a row. We then searched candidate corruptions (byte orders within a beat, row shifts, flips, block reversals, beat permutations and padding variants) for one that reproduces every board output. Only one candidate did: each odd 32-bit input word, four pixels, is replaced by the even word before it ("dup_even"). It reproduces the board on 244 of 244 images and 24 of 24 probes, where the unmodified golden model matches only 12 of the 24 probes. A word pair never crosses a 224-pixel row, which is why the row-constant probes were unaffected.

Prediction before observation. For the second probe set, the outputs predicted by the model were recorded before the board's results were examined. Two single-pixel probes discriminate. For a bright pixel at (1,0), dup_even predicts $(-349, 102, 451)$ for (logit0, logit1, margin), and the golden model predicts the all-zero image's result, $(-375, 106, 481)$. For a bright pixel at (223,223) the roles are exchanged: dup_even predicts the all-zero result and the golden model $(-386, 114, 500)$. The board matched the dup_even predictions exactly.

Mechanism and intervention. The DMA reads DDR through the HP0 port, whose AFI bridge has its own width setting, bit 0 of its RDCHAN_CTRL register [31]. V1.2 used a 32-bit HP0, but PYNQ boots the processing system with the AFI in 64-bit mode and does not run the design's own initialisation code, so each 32-bit read returned the low half of a 64-bit word, the even word. On one boot, the 32-bit bitstream gave 0/244 (run 2), and after that single bit was written from 0 to

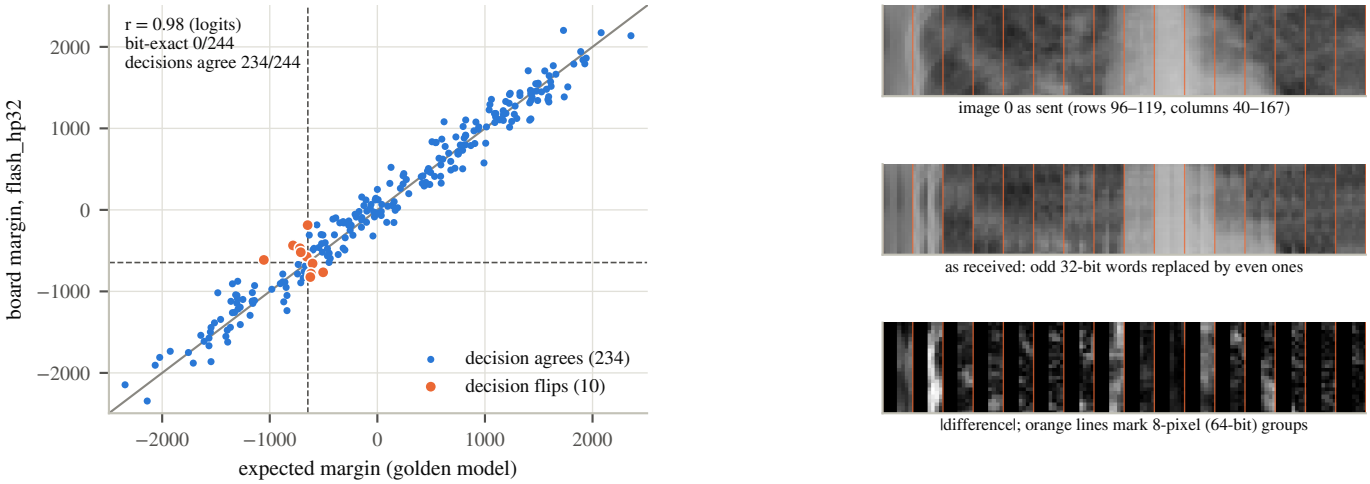


Fig. 6. Left: the first board run, V1.2, against the expected margins (measured). The logits correlate at $r = 0.98$ and 234 of 244 decisions agree, yet no image is bit-exact. Right: a crop of image 0 as sent and as the PL received it under the dup_even model (derived). Each 64-bit group keeps its first four pixels and repeats them, a mild horizontal blur that GAP largely tolerates.

1 it gave 244/244 (run 3). No other setting changed between the two runs, which identifies the register bit as the cause.

Fix. V1.2.1 makes HP0 and the DMA's memory side 64-bit, which matches PYNQ's default and needs no register write (run 1). Every board notebook now compares the AFI setting with the bitstream's hardware description and refuses to run on a mismatch. The simulations drove the stream directly and did not model the processor's memory path, so they could not reveal this fault. Because the network tolerates this kind of input corruption, its decisions stayed mostly correct while every logit was wrong, and the fault was detected only because the board's logits were compared with the reference.

IX. Discussion: Clinical Readiness and the Regulatory Gap

FLASH demonstrates exact hardware execution of the integer network; whether the network is clinically useful is a separate question that this work does not answer. An AUROC of 0.82 on one public adult dataset is research grade. The model separates pneumonia from normal lungs well, but at T its test-set specificity is 33.4% against other abnormalities, compared with 81.3% against normal lungs, and a screener in a real clinic will see such abnormalities constantly. Part of its score comes from the projection, which partly reflects how unwell the patient was when the film was ordered, since AP films are often taken at the bedside. This is the kind of shortcut that [7]–[9] warn about.

Reporting guidelines make the gaps concrete. CLAIM [18] and TRIPOD+AI [19] ask authors to state, among other things, the intended use, how the model was validated, including on external data, and how it performs across subgroups. FLASH reports subgroups and a locked, validation-chosen operating point, but it has no external or paediatric validation, no comparison with radiologists and no prospective data. Pneumonia mortality is concentrated in children [1], while RSNA contains adults.

Threats to validity. The model was trained and tested on one adult dataset from a single source, so its accuracy on other populations, scanners and on children is unknown. The board results cover 244 of the 4,003 test images. The test-set accuracy is the golden model's, and its equality with the board on the remaining images is inferred from the 244. The power figures are vector-less Vivado estimates and have not been measured. The ARM comparison uses unoptimised NumPy code on one core, so the reported speed-up over the ARM is an upper bound on what the FPGA adds.

Regulation adds process on top of evidence. A device would need a software life-cycle process (IEC 62304 [23]), risk management (ISO 14971 [24]) and a quality management system (ISO 13485 [25]). The FDA's action plan [26] describes how AI/ML-based software that changes over time should be managed. Bit-exactness addresses one narrow part of this process, verification. It shows that the deployed hardware computes the validated function on the tested inputs, and a change of threshold is a register value rather than a new model. It does not address clinical validity, and FLASH is therefore a research prototype and not a medical device.

X. Future Work (V2)

The project pauses after V1. Three items would move the remaining software-based results onto the board:

- *A board run on all 4,003 test images.* The 4,003-image accuracy is the golden model's, bit-exact to the board on 244 images; a full board run (about 12 minutes of compute) would make it the board's own output.
- *Measured board power.* The power figures so far are Vivado estimates.
- *DICOM end to end on the board.* The DICOM preprocessing has only been verified on the PC.

Beyond those, the model needs a clinical operating point: explicit treatment of the Not-Normal class, an analysis per

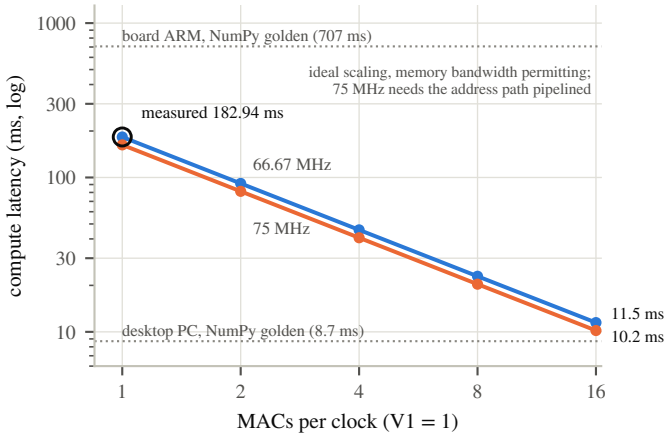


Fig. 7. Projected compute latency against MACs per clock, derived from the measured cycle count. The projection assumes ideal scaling of the MAC cycles, fixed non-MAC cycles and sufficient memory bandwidth; 75 MHz assumes a pipelined address path.

projection, and a threshold chosen on validation data for a stated specificity as well as sensitivity. Since T is a register, none of this needs a hardware change.

On the hardware side, only 14 of 220 DSP slices are used. Several MACs per clock need wider or parallel feature-map reads, since the engine now reads one byte per clock, and a line-buffer window generator is already built and tested for this. Figure 7 projects the gain. At 16 MACs per clock the compute time would be 11.5 ms at 66.67 MHz and 10.2 ms at 75 MHz, still above the desktop PC's 8.7 ms. Any advantage of the FPGA in V2 therefore has to come from power, determinism and cost as well as speed, which is why measured power is among the items above.

XI. Conclusion

FLASH runs a 224×224 integer CNN for pneumonia screening entirely in the programmable logic of a Zynq-7020 and reproduces an independent reference model bit for bit on all 244 verification images, across three bitstreams. A layer-table-driven engine let the same Verilog scale from 28×28 to 224×224 with new memory files and one buffer size. A board-only bus-width fault left 234 of 244 decisions intact while corrupting every logit; logit-level comparison detected it, the golden model localised it offline, and a one-bit intervention confirmed it. The model's accuracy is modest, and its two failure modes, the Not-Normal confusion and the projection shortcut, are reported with it.

Data and Code Availability

The RTL, Vivado scripts, board notebooks, bitstreams, hardware reports, raw board evidence and the scripts that derive every number and figure in this paper are available at https://github.com/Karthik-Flash/ProjectFlash (tag `v1.2.1-hw`). The data are those of the RSNA Pneumonia Detection Challenge on Kaggle [27], described by Shih *et al.* [4], and the radiographs originate from the NIH Chest X-ray dataset [2], [32]. The verification vectors are regenerated from the data by the project's training notebook with a fixed seed.

Ethics and Data Licence

This study used only de-identified, publicly available radiographs and annotations from the RSNA Pneumonia Detection Challenge [4], [27], under the competition's data-use rules and RSNA's terms of use. RSNA's terms permit research use and prohibit any attempt to identify or contact the subjects. The NIH Clinical Center is the provider of the radiographs. No data were collected from patients for this work. The paper reproduces no radiographs; the figures in the repository that show radiographs carry the attribution that the terms require.

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